

Title Page



EXCEL TRANSFER TOOL

TECHNICAL DOCUMENTATION & USER GUIDE



VERSION

1.0 April 2026

Parallel Transfer · Ignore Rows · Cell Selector · Fill All Months

Qt 6

C++17

OpenXML

Multi-Thread

1	Overview	2
2	What’s New in v3.0	3
3	Installation	3
4	User Interface	4
5	Loading Files	5
6	Mapping Cards	7
7	Execute All Transfers	11
8	Rolling Transfer (RT)	11
9	Fill All Months	12
10	Parallel Transfer (Backend)	13
11	Create Month File	14
12	Individual Cell Selector	15
13	Data Flow Reference	16
14	Troubleshooting	18
15	Appendix	19

Excel Transfer Tool is a high-performance Qt/C++ desktop application that automates monthly cost-control reporting workflows. It loads Excel workbooks (XLSM/XLSX) for multiple periods, applies configurable transfer mappings, consolidates data across months, and produces clean output files—all without manual copy-paste work.

Key Capabilities — v1.0

- Load multiple cost-control files with parallel processing (2-3x faster)
- Apply row-level transfer mappings per month/sheet/column
- Rolling Transfer (RT) mode for sequential month-to-month workflows
- Parallel Transfer — run Individual or Rolling modes from a unified backend
- Fill All Months — backfill Jan through target month in one operation
- Ignore Rows — skip specific rows per mapping card
- Create Month File — generate new month files from existing templates
- Visual Cell Selector — pick source/destination cells from a grid view
- Log Panel toggle — show/hide the status log on demand
- Quarter quick-select buttons (Q1–Q4) for rapid period selection
- Automatic formula preservation and shared formula handling

Feature	Description
Parallel Transfer Backend	New <code>ParallelTransferController</code> with <code>IndividualTransferWorker</code> and <code>RollingTransferWorker</code> . Each worker owns its own <code>ExcelHandler</code> and <code>TransferService</code> — fully thread-safe.
Fill All Months	Writes Jan through target month into a single cost-control file. Destination always points to the <code>test/</code> sub-folder.
Ignore Rows Dialog	Per-mapping-card dialog showing all <code>rowMap</code> entries. Check rows to skip during transfer. Filter by row number or value.
Create Month File	Button on each <code>YearCard</code> . Activates creation mode: select months, click Load Periods, app finds last existing file and copies it for missing months.
Visual Cell Selector	“Select Source/Dest Cells” opens a scrollable grid of the loaded Excel sheet. Click any cell to capture its reference.
Log Panel Toggle	“Log” button in header shows/hides the status log dock. Hidden by default on startup.
Copy Full Sheet hidden	For <code>REFI/Budget/prev-year</code> mappings (same-file sources), “Copy full sheet” is automatically hidden.
Run/Remove buttons removed	Mapping row cards no longer show Run and Remove buttons.
Toast positioning fixed	Blue notification toasts now appear at the bottom-center of the application window, not the screen.

System Requirements

Component	Requirement
Operating System	Windows 10 or Windows 11 (64-bit)
Microsoft Excel	Excel 2016, 2019, 2021, or Microsoft 365
Runtime	Qt 6.x (bundled with executable)
Disk Space	Minimum 200 MB free
RAM	8 GB minimum, 16 GB recommended for 12+ months

Setup Steps

1. Extract the application folder to your preferred location.
 2. Ensure `repair.ps1` is in the same folder as `ExcelTransfer.exe`.
 3. Launch `ExcelTransfer.exe`.
 4. Verify the default folder paths point to your network share (L: drive).
 5. Ensure cost-control files follow the expected naming convention.
 6. Verify JSON mapping files exist in the `JSON/` subfolder.



Header Bar

The header bar contains global controls:

Button	Action
Reset	Clears all loaded files, mapping cards, and period rows
Create Month File	Opens creation mode on the year cards (see Section 11)
Log	Toggles the status log panel (hidden by default)

Year Cards

Each year card shows month checkboxes and quick-select controls:

- **Year checkbox** — expands/collapses the card
- **Q1–Q4 buttons** — select/deselect entire quarters
- **All / None** — select or deselect all 12 months
- **+ Create Files** — activates file creation mode (amber button, appears when card is expanded)

Period Limit

Maximum **5 months** can be selected at a time for Execute All. Selecting more shows a warning and prevents execution to avoid memory overflow.

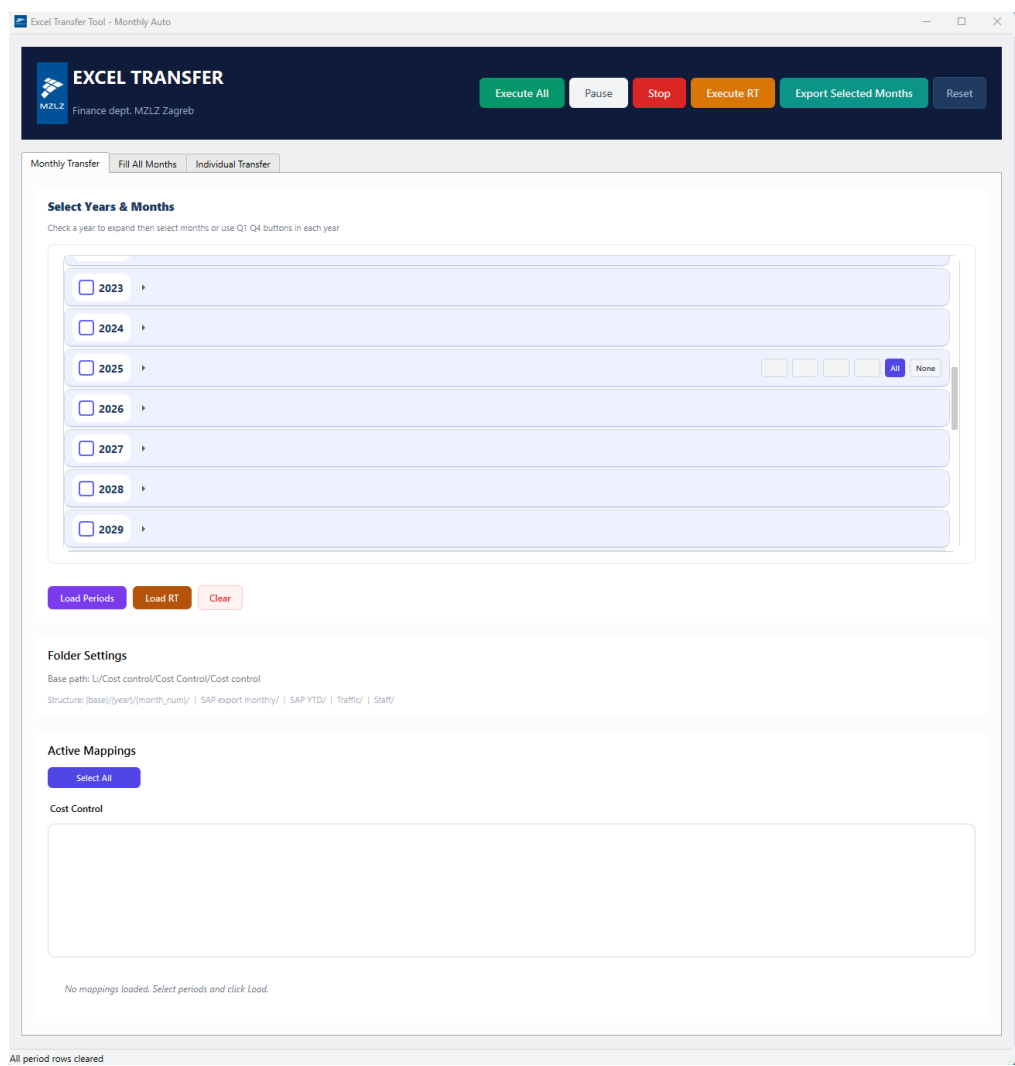


Figure 1: Excel Transfer Tool — Main window showing the Monthly Transfer tab with collapsed year cards.

Log Panel

Click the **Log** button in the header to show or hide the status log dock. The log panel is hidden by default on startup. When visible, it shows all transfer status, warnings, and debug output in real time.

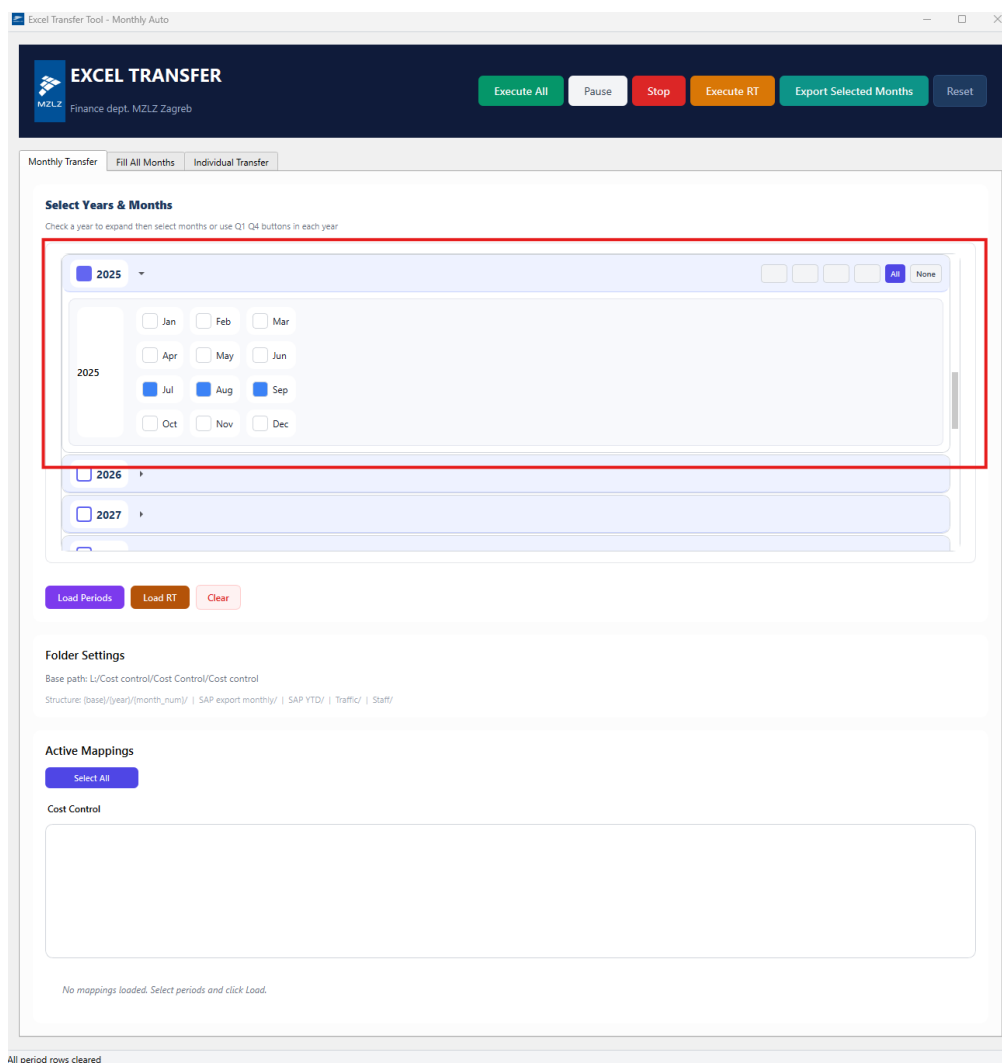


Figure 2: Year card expanded showing month checkboxes. July, August and September are selected (blue). Q1–Q4, All, and None quick-select buttons are visible on the right.

Selecting Periods

1. Check the year checkbox (e.g. 2025) to expand the year card.
2. Select months individually or use Q1–Q4 / All / None buttons.
3. Click **LOAD Selected Periods**.
4. Wait for the green “Loaded” toast notification.

Important

Ensure all target files are **closed in Excel** before loading. Open files cause ZIP read errors.

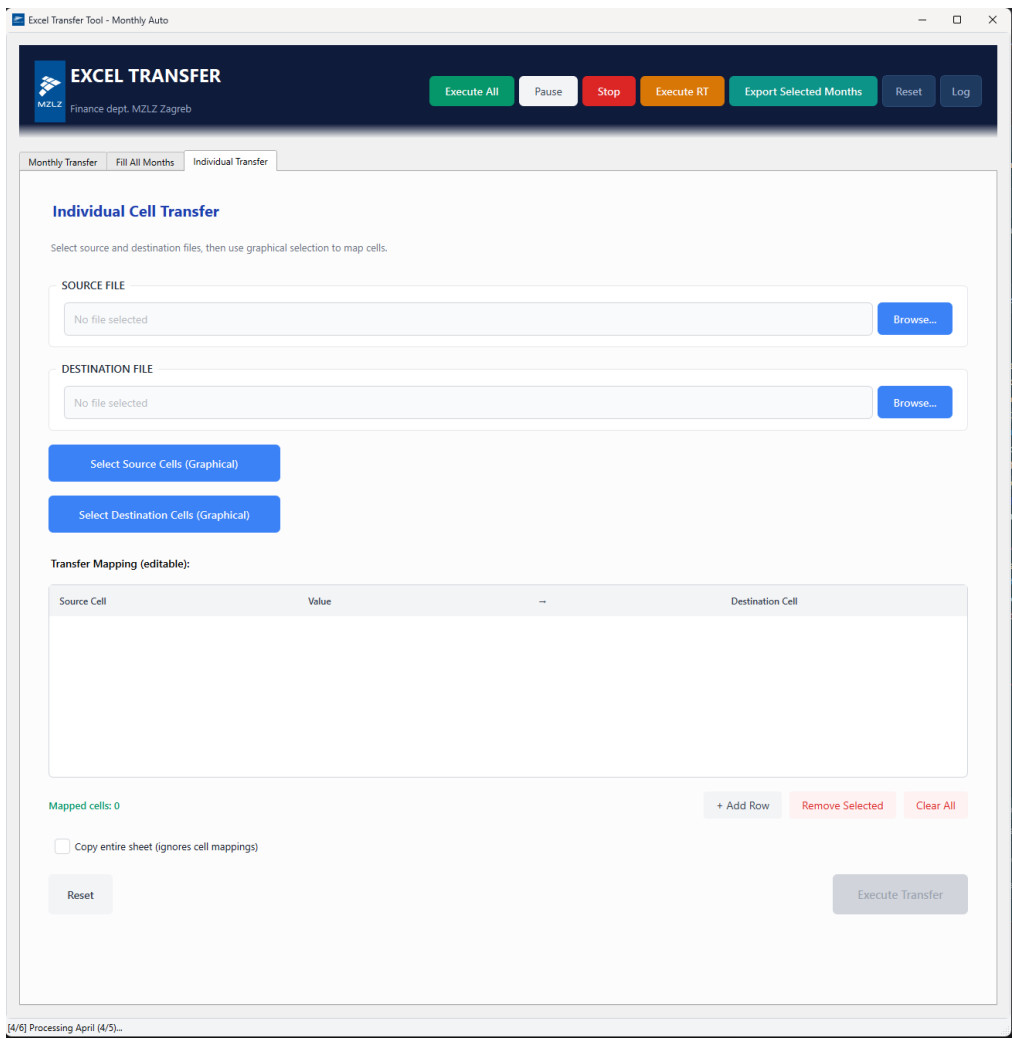


Figure 3: Individual Transfer tab showing the **Log** button in the header bar (top right). The status bar at the bottom displays live transfer progress.

What is a Mapping Card?

Each mapping card represents a transfer rule:

- A **source** workbook, sheet, and column
- A **destination** sheet and column
- A **rowMap** defining source row → destination row pairs

Mapping Card Buttons

Button	Action
Edit Rows	Modify source/destination row numbers
Export	Export rowMap to JSON file
Import	Import rowMap from JSON file
 Ignore	Open Ignore Rows dialog to skip specific rows

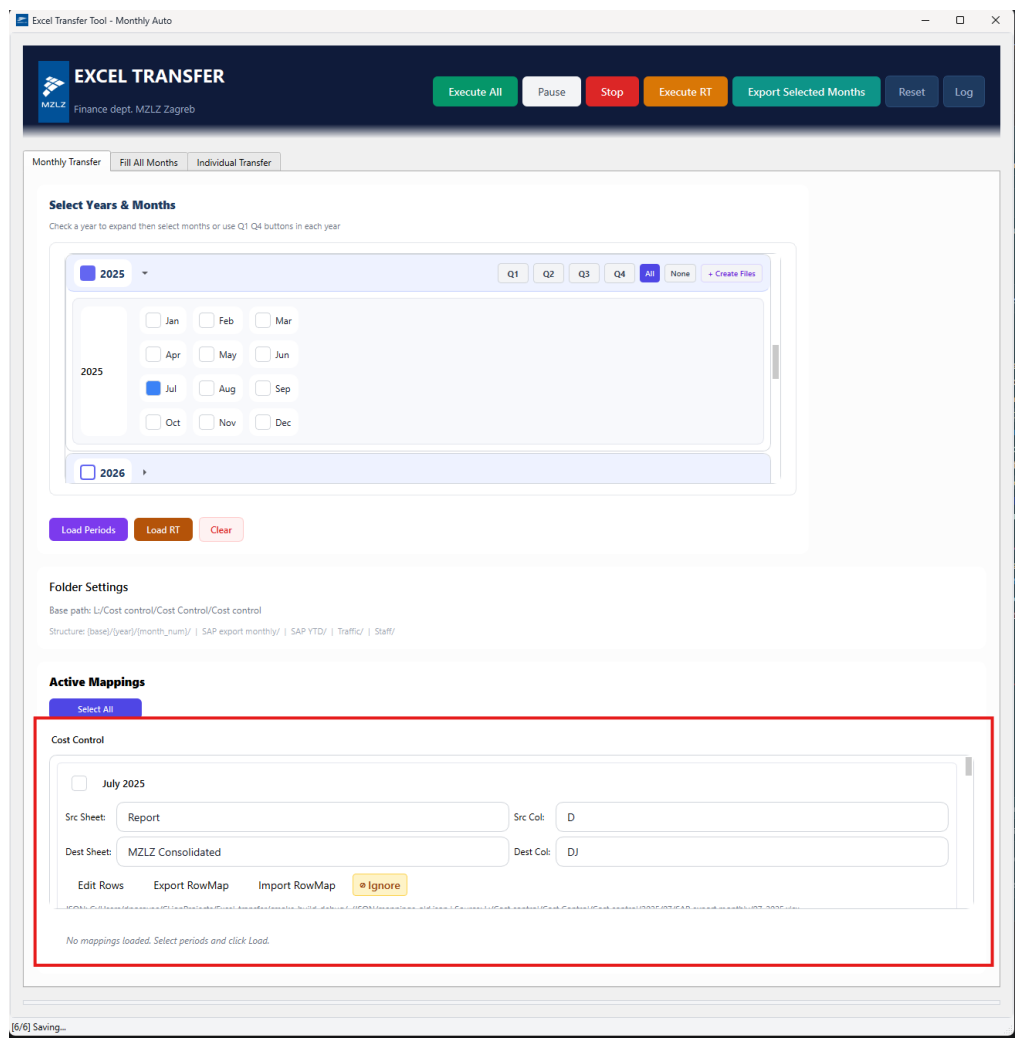


Figure 4: Active Mappings section showing a loaded mapping card (SAP Report → MZLZ Consolidated). The card shows Src/Dest Sheet, Src/Dest Column, and action buttons: **Edit Rows**, **Export RowMap**, **Import RowMap**, and **Ignore**. The + **Create Files** button is visible on the year card header.

Copy Full Sheet

The “Copy full sheet” checkbox is **automatically hidden** for REFI, Budget, and prev-year mappings. These mappings read from the same cost-control file — copying a sheet to itself is not applicable.

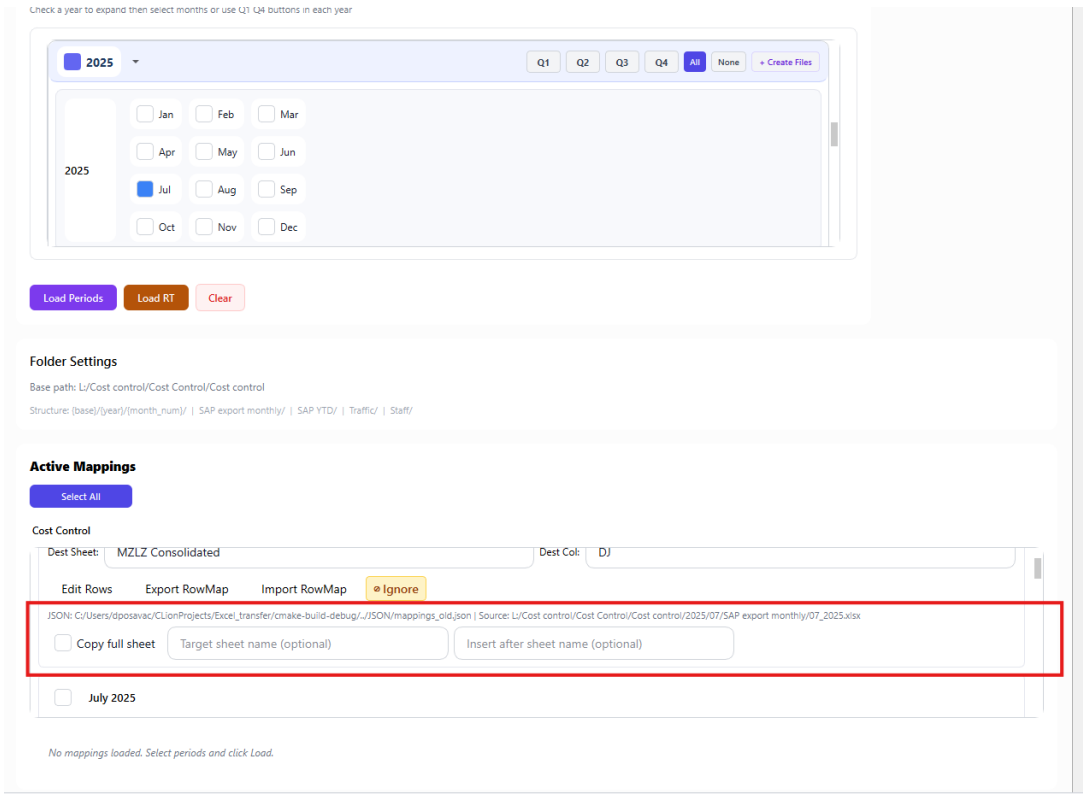


Figure 5: **Copy Full Sheet** option on a SAP mapping card. When enabled, the entire source sheet is copied to the destination workbook. Two optional fields allow specifying a **Target sheet name** (rename on copy) and **Insert after sheet name** (position in workbook). This option is hidden for REFI/Budget/prev-year mappings since those sheets already exist in the destination file.

Ignore Rows Dialog

Click ð **Ignore** on any mapping card to open the Ignore Rows dialog:

1. The dialog shows a table of all rowMap entries: Dest Row, Dest Column, Src Row(s), Current Value.
2. Check the “Skip” checkbox next to any row you want to exclude from transfer.
3. Use the **Filter** box to search by row number or cell value.
4. Click **Ignore All** / **Ignore None** for bulk selection.
5. Click **Apply** to save. The status shows how many rows will be skipped.

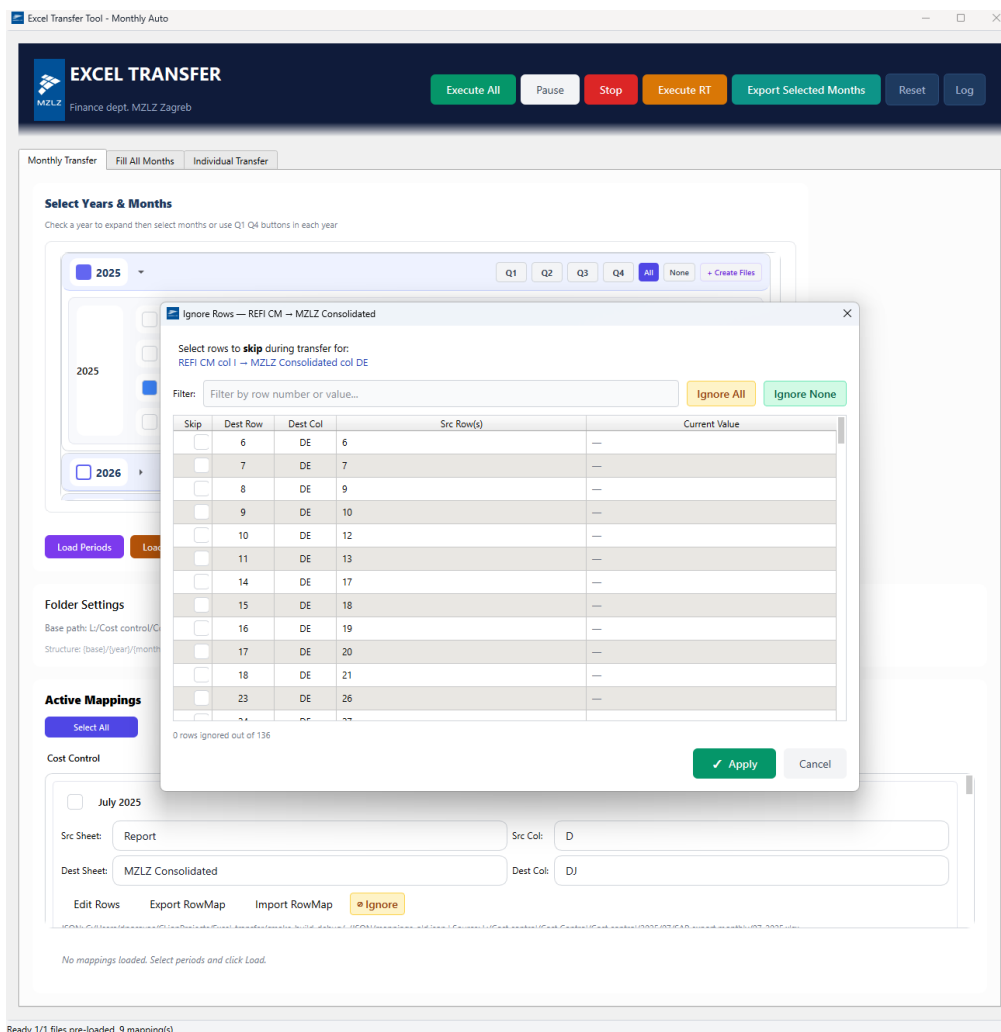


Figure 6: Ignore Rows dialog for REFI CM → MZLZ Consolidated. Each rowMap entry shows Dest Row, Dest Column, Src Row(s), and Current Value. Check **Skip** to exclude a row from the next transfer. The filter box narrows the list. Status line shows “0 rows ignored out of 136”.

Session Only

Ignored rows are stored in memory for the current session. They are not persisted to JSON files and will reset when you restart the application.

1. Ensure files are loaded (green status).
2. Check the mapping cards you want to execute.
3. Click **Execute All**.
4. Monitor progress. On completion, a summary shows total cells transferred.

Maximum 5 Months

Selecting more than 5 months simultaneously will show a warning and block execution. Process months in batches of up to 5 to avoid memory overflow.

Pause and Stop

- **Pause** — temporarily halts execution (can be resumed)
- **Stop** — terminates execution (current sheet finishes first)

What is Rolling Transfer?

Rolling Transfer processes months **sequentially**, where each month's output becomes the input for the next. Use RT when building month-to-month cumulative data chains.

How RT Works

1. Select multiple months in chronological order.
2. Click **Load RT** to validate source files exist.
3. Click **Execute RT**:
 - Copies previous month file to new month's **test/** folder
 - Applies all mappings (traffic_mott, pax_transfer, pax, staff, SAP, REFI, sap_ytd)
 - Saves and reloads before proceeding to next month

RT Error Handling

RT **aborts** on the first error. Because each month depends on the previous output, continuing after an error would produce incorrect cumulative data.

What is Fill All Months?

Fill All Months writes data for **all months January through a target month** into a single cost-control file. Use this to backfill an entire year’s worth of data into one destination file.

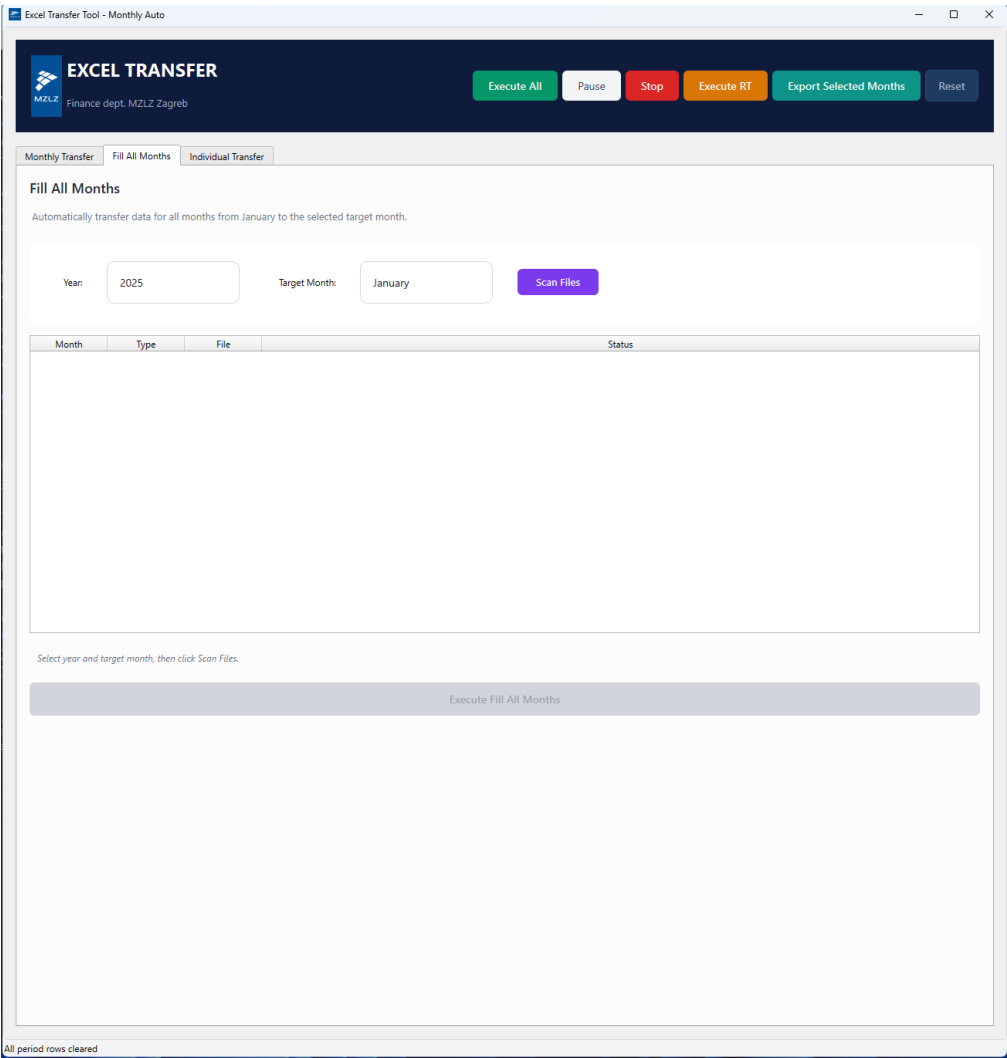


Figure 7: Fill All Months tab — select year and target month, then click Scan Files to discover all source files.

How it Works

1. Navigate to the **Fill All Months** tab.
2. Select the **year** and **target month** (e.g. November 2025).
3. Click **Scan Files** — the app scans for all source files (SAP, PAX, Staff, etc.) for each month Jan through November.
4. Review the scan table: green checkmarks show found files, red X marks show missing files.
5. Click **Execute Fill All Months** to run the transfer.

Destination Path

The destination file is always in the `test/` subfolder:
`L:/Cost control/.../2025/11/test/Cost Control ZAG 11_2025_working.xlsm`

Corrupt Files

If the destination file shows “not a zip file” error, it may be corrupt. Replace it with a fresh copy from the backup folder before running Fill All Months.

Overview

The Parallel Transfer system is a new backend that provides two independent transfer modes, each running on its own thread with its own `ExcelHandler` and `TransferService` instances:

Mode	Behaviour
Individual	Each month runs independently. On error: logs and continues to next month. No save+reload between months.
Rolling	Months run sequentially in calendar order. On error: aborts remaining months (data integrity). Save+reload after each month so next month reads fresh output.

Key Design Principles

- **Thread safety** — each worker owns its own handler/service, never shared
- **Decoupled** — deleting the parallel transfer files does not affect existing functionality

- **Validation** — Rolling requires ≥ 2 contiguous months before thread is created
- **Cancellable** — both workers check `m_cancelled` flag between entries

Rolling Validation Rules

The Rolling worker validates before starting:

- At least **2 months** must be selected
- Months must be **contiguous** (no gaps, e.g. Jan + Mar without Feb is rejected)
- Validation runs on the **UI thread** (instant feedback) and again on the **worker thread** (safety)

How it Works

The Create Month File feature copies an existing cost-control xlsx as a template for new months:

1. Expand a year card (e.g. 2025).
2. Click the amber + **Create Files** button.
3. Select the months you want to create (e.g. May, June, July).
4. Click **Load Periods** (now in creation mode).
5. The app automatically finds the **last existing** cost-control file for that year (scans December → January).
6. A confirmation dialog shows the source file and destination paths.
7. Click **Yes** to create the files.

Source File Selection

The app always copies from the **most recent existing** month. If only February exists, it uses February for March, April, May, etc. Files are created in the `test/` subfolder of each month's directory.

Result

A toast notification confirms: "Created 3 file(s) from February 2025." Already-existing files are skipped (not overwritten).

When configuring an **Individual Transfer**, click **Select Source Cells** or **Select Destination Cells** to open a visual cell picker:

- **Sheet dropdown** at the top — switch between sheets
- **Scrollable grid** showing cell data (rows 1–50, columns A–Z)
- **Click any cell** — automatically fills the Column and Row fields
- **“Selected:” label** updates in real time (e.g. `Sheet1!N17`)
- **Manual override** — type column letter and row number directly
- Click **Apply** to confirm selection

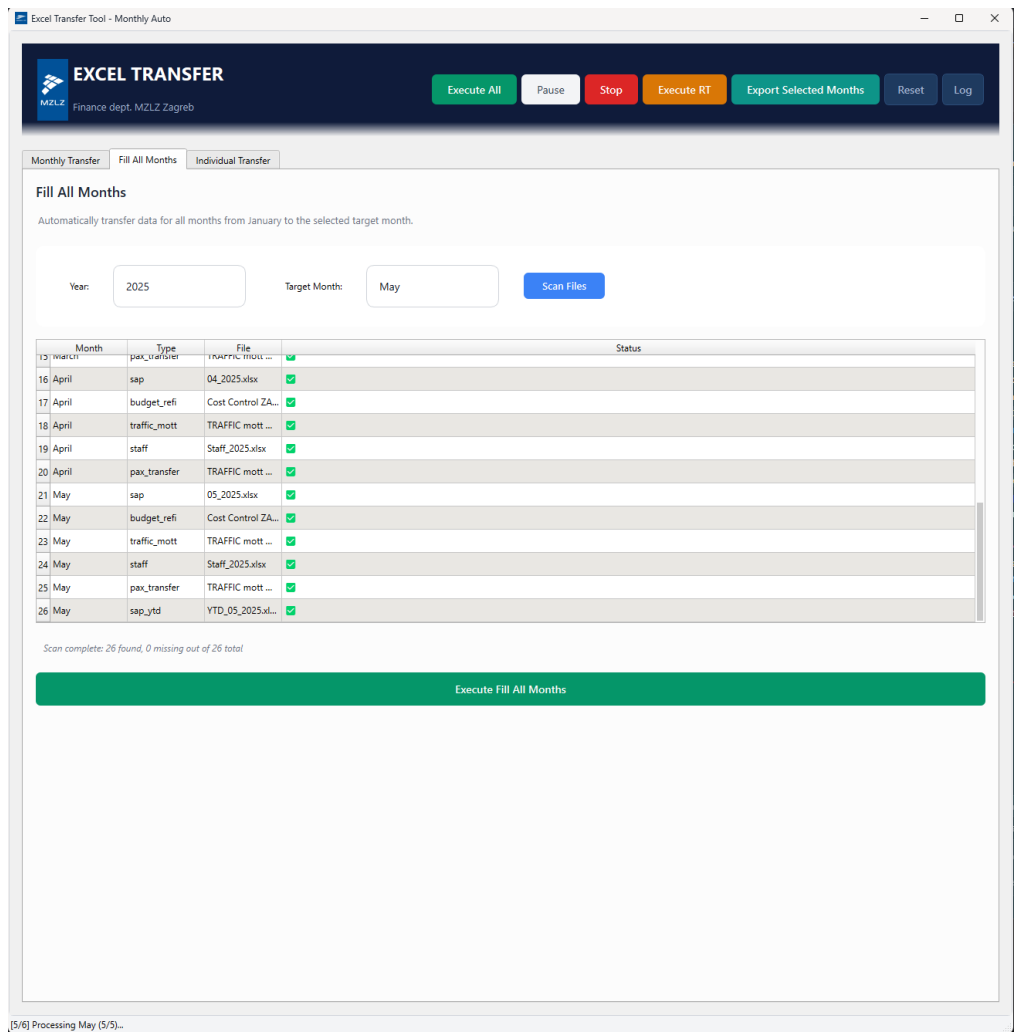


Figure 8: Individual Transfer tab — browse source and destination files, use graphical cell selection buttons, and configure the transfer mapping table.

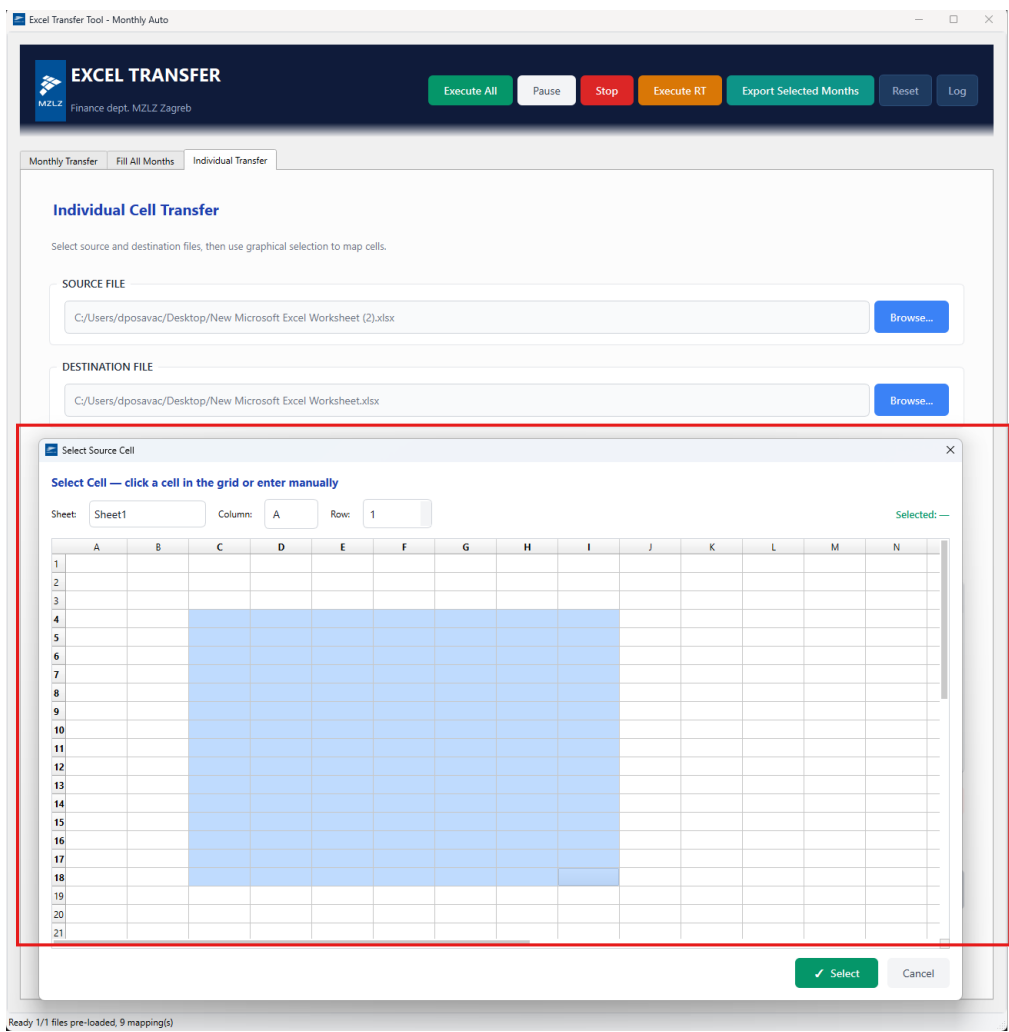


Figure 9: Visual Cell Selector dialog — “Select Source Cell”. The grid shows the loaded workbook data (Sheet1, columns A–N, rows 1–21). Click any cell to capture its reference. The Sheet, Column, and Row fields update automatically. The highlighted area shows existing data in the source file.

File Must Be Loaded

The cell selector reads data from the already-loaded workbook. If the source or destination file has not been loaded, the grid will be empty. Load the file first using Load Periods.

Excel Transfer

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Execute All

Pause

Stop

Execute RT

Export Selected Months

Reset

Log

Monthly Transfer

Fill All Months

Individual Transfer

Individual Cell Transfer

Select source and destination files, then use graphical selection to map cells.

SOURCE FILE

C:/Users/dposavac/Desktop/New Microsoft Excel Worksheet (2).xlsx

Browse...

DESTINATION FILE

C:/Users/dposavac/Desktop/New Microsoft Excel Worksheet.xlsx

Browse...

Select Source Cells (Graphical)

Select Destination Cells (Graphical)

Transfer Mapping (editable):

	Source Cell	Value	→	Destination Cell
1	D4	0.00	→	
2	C4	0.00	→	
3	I4	0.00	→	
4	H4	0.00	→	
5	F4	0.00	→	
6	E4	0.00	→	
7	G4	0.00	→	

Mapped cells: 105

+ Add Row

Remove Selected

Clear All

☐ Copy entire sheet (ignores cell mappings)

Reset

Execute Transfer

Figure 10: Individual Transfer tab with 105 cells mapped. The Transfer Mapping table lists source cells (D4, C4, I4, H4, F4, E4, G4) with their current values and destination cells. The **Execute Transfer** button becomes active once mappings are configured.

Traffic Mott YTD Flow

Step	Source	Destination
1	External TRAFFIC mott YYYY.xlsx Sheet1	Cost control TRAF- FIC mott YYYY sheet (cols D–O)
2	External TRAFFIC mott YYYY.xlsx Sheet1	Cost control TRAFFIC mott Q (Q9/Q10 month number)
3	External TRAFFIC mott YYYY.xlsx (sum G..currentCol)	MZLZ Consoli- dated JG rows 5,7,8,9,10,11
4	Staff {year} FTE sheet row 33	MZLZ Consoli- dated JG row 16

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Page 17 of 21

YTD Calculation

PAX source columns G–R contain **single-month values** (not cumulative). The JG column YTD is computed in C++ by summing columns G through the current month column directly from the external TRAFFIC mott file.

Transfer Execution Order

Mappings execute in this fixed priority order:

Priority	Type	What it does
1	traffic_mott	Writes monthly columns D–O in TRAFFIC mott sheet
2	pax_transfer	Writes PAX values to TRAFFIC mott (same sheet)
3	pax	Writes PAX monthly values to MZLZ (col GX etc.)
4	staff	Writes FTE data from Staff file
5	sap	Writes SAP monthly data to MZLZ
6	budget_refi	Writes REFI, Budget, prev-year columns
7	sap_ytd	LAST: reads Q from TRAFFIC mott, writes JG + FTE

“not a zip file” Error

Symptom

QZip: not a zip file! in log. All 9 mappings fail with 0 cells.

Cause: The cost-control xlsx file is corrupt or was created from a corrupted template.

Fix: Replace the file with a fresh copy from the Backup folder or copy from another month’s working file.

0 Cells Transferred

- Verify destination file is in the test/ subfolder
- Check source files (SAP, PAX, Staff) are present and correctly named
- Ensure no rows are accidentally marked as “Ignore” in the Ignore Rows dialog
- Review log panel for “sheet not found” or “file missing” warnings

Stack Overflow / Crash on Execute All

Symptom

Application crashes with exit code 0xC0000602 when clicking Execute All.

Cause: Too many months selected simultaneously creates too many Qt widgets.

Fix: Select **maximum 5 months** at a time. The app will show a warning if more are selected.

Fill All Months Writes to Wrong Folder

Ensure the destination path shown in the scan table ends with `/test/`. If it points to the root month folder, the file naming in that folder may be non-standard.

Parallel Transfer Not Visible in UI

The Parallel Transfer backend is implemented but the UI tab is under development. The backend can be connected by wiring `ParallelTransferController` signals in `mainwindow.cpp`.

File Naming Convention

```
Cost Control ZAG {MM}_{YYYY}_working.xlsm
Location: L:/Cost control/Cost Control/Cost control/{YYYY}/{MM}/test/
```

File Type	Path Pattern
Cost Control	.../{YYYY}/{MM}/test/Cost Control ZAG {MM}_{YYYY}_working.xlsm
SAP Monthly	.../{YYYY}/{MM}/SAP export monthly/{MM}_{YYYY}.xlsx
SAP YTD	.../{YYYY}/{MM}/SAP export monthly/YTD_{MM}_{YYYY}.xlsm
Traffic (PAX)	.../{YYYY}/{MM}/Traffic/TRAFFIC mott {YYYY}.xlsx
Staff	.../Staff/Staff_{YYYY}.xlsx

JSON Mapping Files

File	Type	Purpose
mappings_old.json	sap	SAP monthly → MZLZ Consolidated
mappings_budget_refi_prev_year.json	budget_refi	REFI/Budget/PrevYear → MZLZ
Traffic_mott.json	traffic_mott	PAX Sheet1 → TRAF-FIC mott sheet
mappings_pax_transfer.json	pax_transfer	PAX Sheet1 → TRAF-FIC mott (extra rows)
mappings_sap_ytd.json	sap_ytd	TRAFFIC mott Q → MZLZ JG
pax.json	pax	PAX Sheet1 → MZLZ monthly PAX cols
staff.json	staff	Staff FTE → MZLZ

Technical Architecture

Component	Technology
UI Framework	Qt 6.x (Widgets)
Language	C++17
Excel Parsing	OpenXML (ZIP + XML), no COM dependency
Compression	miniz (embedded)
Threading	QThread, QThreadPool
JSON Parsing	Qt JSON (QJsonDocument)
Parallel Transfer	ParallelTransferController + IndividualTransferWorker + Rolling-TransferWorker

Performance Characteristics

Operation	Performance
Transfer 1 month (all 9 mappings)	10–60 seconds (depends on file size)
Save 50MB XLSM file	8–20 seconds
Fill All Months (12 months)	3–8 minutes
RT chain (6 months)	2–4 minutes
Maximum safe months (Execute All)	5 months

Excel Transfer Tool v1.0

• User Manual

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